10 HEXA-MOD

Modular hex pads on four height levels, colour-coded (red = tallest).

66.1 g

PLA · 100 % infill · sealed core

✓ LIGHTENED CORE · SLOT + HOLE



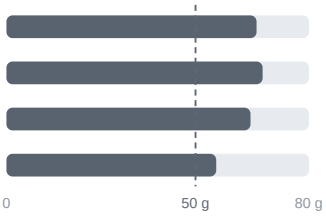
3/4 view · rendered from the STL, colours are preview only

AT A GLANCE

Overall height	**135.00 mm**
Max footprint (foot)	**42.63 × 36.61 mm**
Paddle slot	**30.2 × 12.2 mm, 130.0 deep**
Textured band	**z 21.5 – 116.0 mm (94.5 tall)**
Outer size in band (max)	**38.7 × 33.0 mm**
Peak texture height	**3.50 mm above base**
Screw hole (butt)	**Ø 3.5 mm + Ø 6.2 × 2.2 counterbore**
Material volume / surface	**53.34 cm³ / 459 cm²**

WEIGHT · 100 % INFILL

PLA	1.24 g/cm³	**66.1 g**
PETG	1.27 g/cm³	**67.7 g**
TPU	1.21 g/cm³	**64.5 g**
ABS	1.04 g/cm³	**55.5 g**



COLOURWAY



#1C1C1E · #464A4E · #74787E · #B0B4BA · #D63A28

3MF print palette

PRINT PROFILE

Infill / layer height	**100 % / 0.20 mm**
Layers	**675**
Orientation	**upright, butt end down**
Supports	**none required**
Sealed hollow core	**34.0 cm³, walls ≈ 1.2 mm**
Filament, 1.75 mm	**≈ 22.2 m (100 % flow)**

DESIGN INTENT

A honeycomb of individual hex pads, each separated by a 1.7 mm gap and set at one of four heights. The height map is pseudo-random with a gentle swell along the handle; the tallest pads (red) reach the full 3.5 mm crest.

Pads are chamfered on the lower edge so they never overhang beyond 45 degrees.

KEY FEATURES

- 12 columns × 11 rows of pads, 10.2 mm across flats
- Four levels: 2.35 / 2.76 / 3.13 / 3.50 mm above the base
- Red pads mark the tallest level (about 6% of the band)

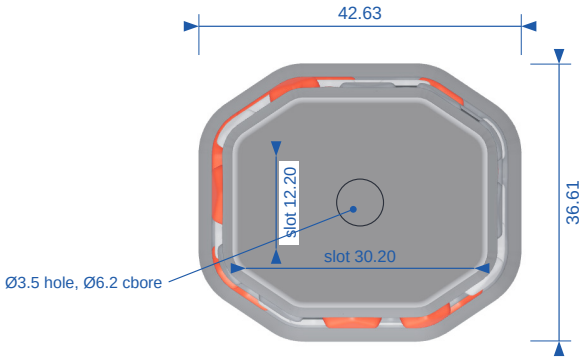
VERIFIED ON THE STL

Paddle slot	**30.2 × 12.2 mm (30 × 12 + 0.2)**
Screw hole + counterbore	**Ø 3.5 mm + Ø 6.2 × 2.2**
Outer envelope vs reference	**+0.000 mm at most**
Outer body / sealed cavity	**1 / 1**
Wall around the cavity (min)	**1.09 mm**
Material under 0.4 mm wide	**0.005 % of volume**
Widest flat bridge	**3.4 mm**

✓ SLOT & HOLE VERIFIED ✓ ONE BODY · SEALED CORE ✓ PRINT-READY · 0.20 MM LAYERS ✓ 3MF VALIDATED

FILES

stl/10_hexa_mod.stl	17.5 MB · 350,420 triangles
3mf/10_hexa_mod.3mf	3.7 MB · colour

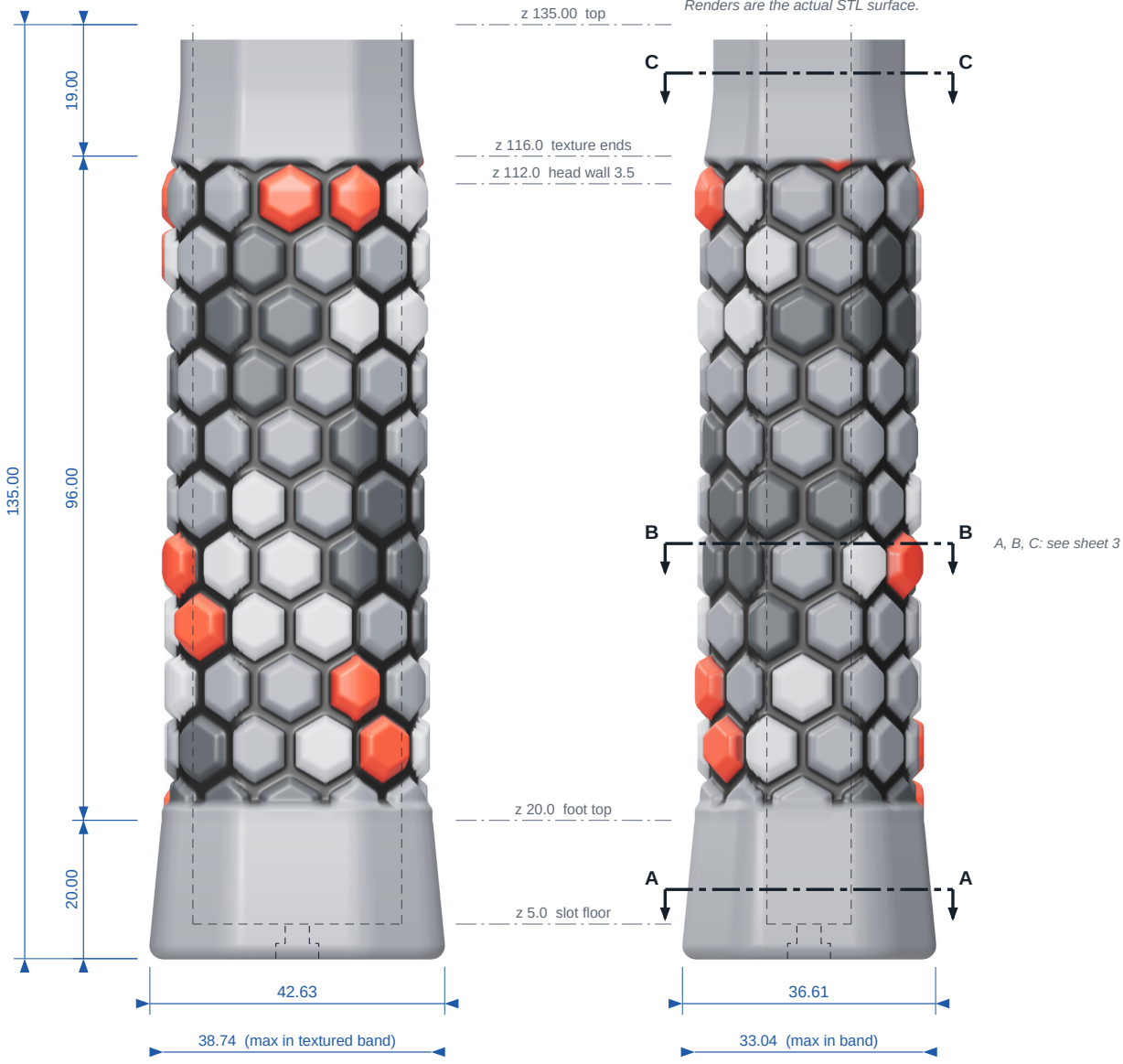


TOP VIEW (from +Z)

KEY DIMENSIONS

Overall height	**135.00**
Overall width × depth (foot)	**42.63 × 36.61**
Paddle slot width × depth	**30.20 × 12.20**
Slot depth (floor at z 5.0)	**130.00**
Outer W × D, textured band (max)	**38.74 × 33.04**
Foot top (z 20)	**39.16 × 33.13**
Butt face (z 0)	**40.56 × 34.54**
Peak texture height above base	**3.50**
Screw hole / counterbore	**Ø 3.50 / Ø 6.20 × 2.2**

All values in mm. Dashed lines: paddle slot and screw hole (hidden edges). Renders are the actual STL surface.



A, B, C: see sheet 3

FRONT VIEW (from −Y)

RIGHT VIEW (from +X)

SCALE 1 : 1



PART

HEXA-MOD

Pickleball paddle grip sleeve, closed butt, open throat

GENERAL NOTES

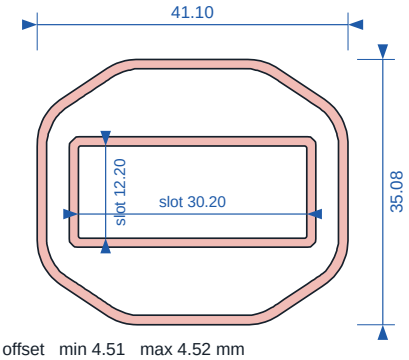
1. Views are true size: print this sheet at 100 %, do not fit to page.
2. Dimensions are measured from the STL, not nominal.
3. Sealed hollow core, walls ≈ 1.2 mm; head above z 100 solid, end walls slit 20 mm.
4. Outer envelope matches PickleballGrip_1.3mf below z 90; head thickened.

SCALE **1 : 1**	UNITS **millimetres**	PROJECTION
MATERIAL **PLA / PETG / TPU**	INFILL **100 %**	LAYER HEIGHT **0.20 mm**
MASS (PLA) **66.1 g**	VOLUME **53.34 cm³**	SURFACE AREA **459 cm²**
REVISION **7**	DATE **2026-10-22**	SHEET **2 of 4**

Sections & profiles

True-size cross-sections, how the outer surface and material change with height, and the dimension tables.

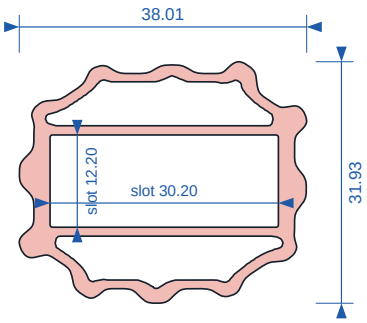
SECTION A–A z = 10 mm · foot



Skin offset min 4.51 max 4.52 mm

Area 268 mm² · outer perimeter 128.9 mm

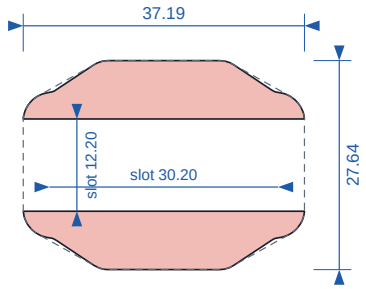
SECTION B–B z = 60 mm · textured band



Skin offset min 1.20 max 3.48 mm

Area 295 mm² · outer perimeter 117.7 mm

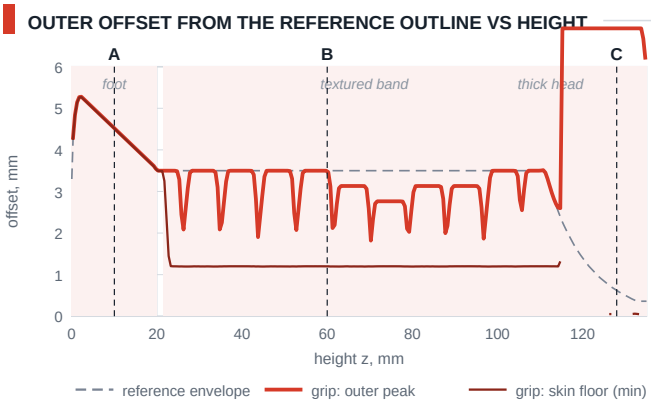
SECTION C–C z = 128 mm · rim



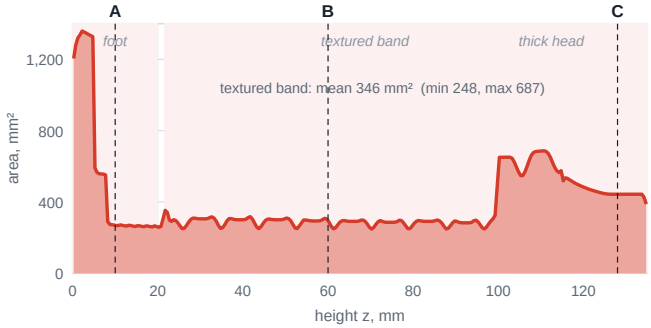
Skin offset max 6.92 mm

Area 443 mm² · outer perimeter 110.7 mm

OUTER OFFSET FROM THE REFERENCE OUTLINE VS HEIGHT



MATERIAL CROSS-SECTION AREA VS HEIGHT



OVERALL & SLOT

Overall length (Z)	**135.00 mm**
Overall width (X)	**42.63 mm**
Overall depth (Y)	**36.61 mm**
Slot width × depth	**30.20 × 12.20 mm**
Slot area / perimeter	**368.4 mm² / 84.3 mm**
Slot depth (floor at z 5.0)	**130.00 mm**
Screw hole / counterbore (butt)	**Ø 3.50 / Ø 6.20 × 2.2**
Sealed cavity / min wall	**34.0 cm³ / 1.09 mm**
Outer W × D in band (max)	**38.74 × 33.04 mm**
Material volume / surface	**53.34 cm³ / 459.5 cm²**

ZONES BY HEIGHT

ZONE	Z FROM	Z TO	HEIGHT
Solid floor (butt)	0.0	5.0	**5.0**
Flared foot	0.0	20.0	**20.0**
Smooth collar	20.0	21.5	**1.5**
Textured band	21.5	116.0	**94.5**
Smooth neck	116.0	112.0	**-4.0**
Head (solid, thick wall, slot opening)	112.0	135.00	**23.0**

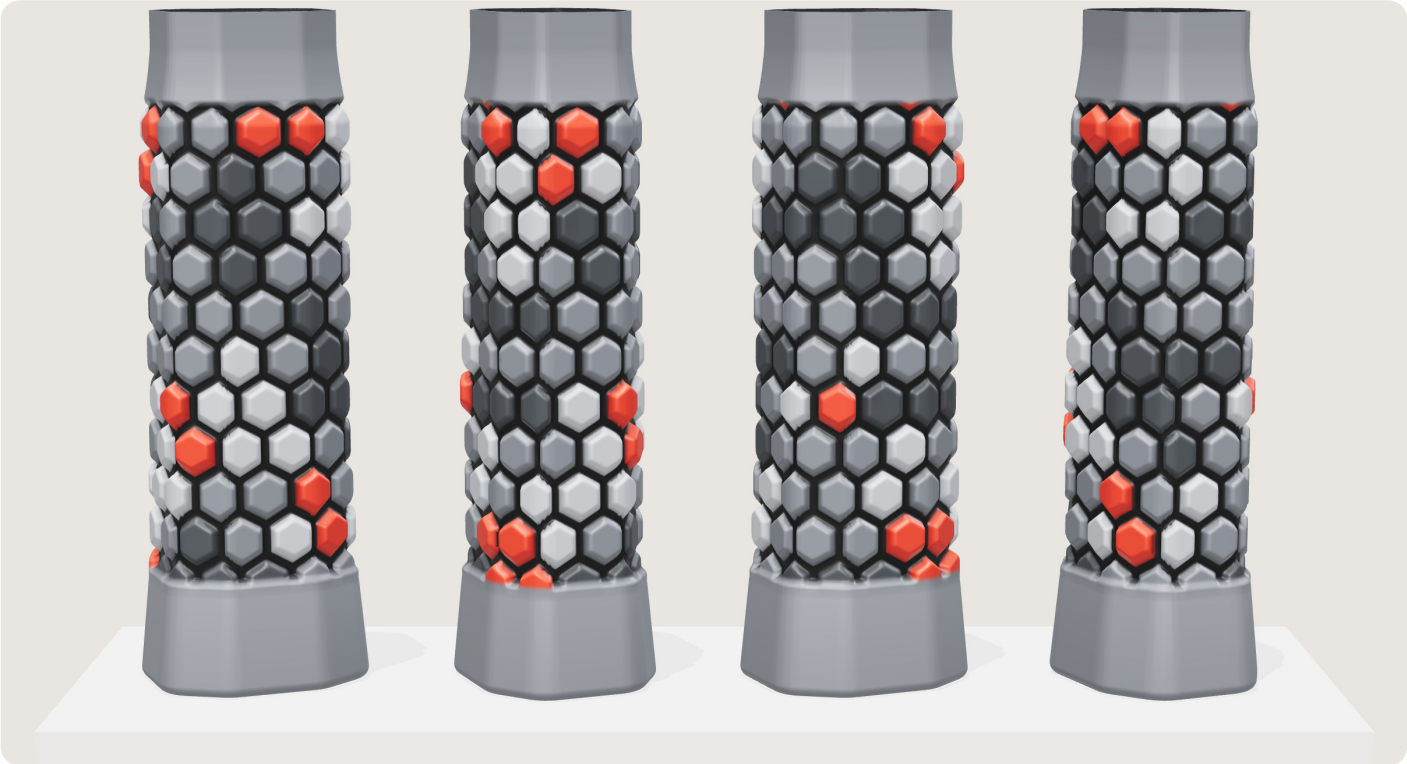
Relief floor above reference line (design / min)	**1.20 / 1.19 mm**
Slot wall over the head (design / min)	**3.50 / 3.50 mm**
Peak texture height above base	**3.50 mm**

OUTER DIMENSIONS BY HEIGHT

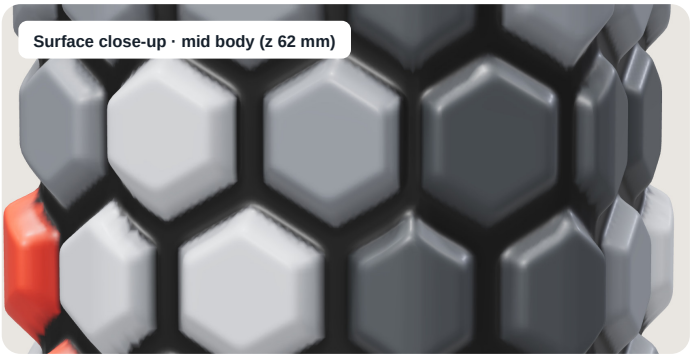
Z (MM)	OUTER W	OUTER D	PERIMETER	AREA MM²	SKIN MIN	PEAK OFFSET
5.0	42.08	36.06	132.0	959	5.00	5.01
10.0	41.10	35.08	128.9	269	4.51	4.52
20.0	39.11	33.09	122.6	260	3.52	3.52
30.0	38.33	32.67	118.8	305	1.20	3.50
45.0	37.80	32.03	115.1	287	1.20	3.13
60.0	38.07	31.91	117.6	294	1.20	3.48
75.0	37.59	31.49	117.0	286	1.20	2.76
90.0	37.15	32.20	117.7	292	1.20	3.13
105.0	38.03	31.09	113.3	558	1.20	3.08
116.0	37.20	30.24	114.1	534	open	6.92
125.0	37.19	27.75	110.8	447	open	6.92
132.0	37.19	27.64	110.7	443	0.06	6.92

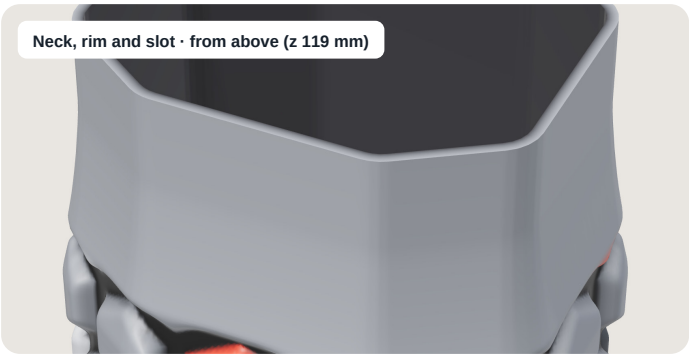
Views & surface

Four-way turntable, close-ups of the surface, and the unrolled height map with the design parameters.

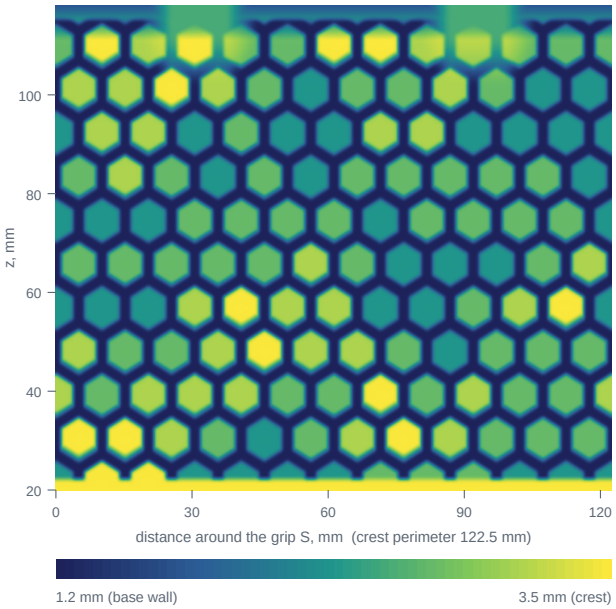


Four views, 90° apart · rendered from the STL, colours are preview only





UNROLLED SURFACE · HEIGHT ABOVE BASE



DESIGN PARAMETERS

Pad grid	**12 × 11 (pointy-top hex)**
Pad across flats	**10.21 mm**
Gap between pads	**1.7 mm**
Levels	**2.35 / 2.76 / 3.13 / 3.50 mm**
Gap floor	**1.20 mm (base wall)**
Pad edge	**chamfered, about 0.8 mm**
Level colours	**3 greys + red**

PRINT & VERIFICATION

Result: slot & hole / print-ready	**PASS / PASS**
Outer body · genus · sealed cavity	**1 · 1 · 1**
Wall around the cavity (min)	**1.09 mm**
Paddle slot	**30.2 × 12.2 mm**
Envelope over reference	**+0.000 mm**
Material under 0.4 mm wide	**0.005 %**
Widest flat bridge	**3.4 mm**